Priyanshu Debnath

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B.Tech - Electronics and Electrical Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	9.68 (Current)	2025-Present
B.Tech. Minor	Indian Institute of Technology, Guwahati	0.0 (Current)	2026-Present
Senior Secondary	WBCHSE Board	95.0%	2025

Experience

- **Coding Club, IIT Guwahati** *Jan. 2026 - Present*
  Deputy Coordinator Guwahati, India
  - Lead the Competitive Programming Module; design problem sets, conduct algorithm training sessions, and organize inter-campus coding contests
- **Finance & Economics Club, IIT Guwahati** *Jan. 2026 - Present*
  Quant Associate Guwahati, India
  - Work on quantitative finance projects spanning Risk Modeling and Stochastic Calculus; research financial derivatives pricing and trading strategy optimization

Projects

- **BDH-ECHR: Post-Transformer Interpretable Legal AI** *Jan. 2026 – Feb. 2026*
  Python, PyTorch, Dragon Hatchling (BDH), Hebbian Learning, Gemini 2.0 Flash Github
  - Applied Dragon Hatchling post-transformer to multi-label ECHR violation classification; achieved **Micro-F1 = 0.7716**, outperforming BERT-base (+13.2 pts) and LegalBERT (+5.2 pts) at **5.4× fewer parameters** (20.4M vs. 110M)
  - Engineered hierarchical chunking (3,584-token context, 7× BERT) and conducted the first **neuron→concept monosemanticity audit**: 218 neurons mapped to 477 legal concepts; 2 with perfect isolation (E=1.0); per-doc sparsity 0.305% — 16× below BDH's own target
- **SpeakX Aurora — Self-Learning Notification Orchestrator** *Jan. 2026 – Feb. 2026*
  Python, UMAP, Scikit-learn, Groq LLaMA-3.3-70B, LangChain, Pandas Github
  - Domain-agnostic ML pipeline (Kriti 2026): **UMAP + Agglomerative Clustering** for MECE behavioral segmentation; Octalysis-driven bilingual (EN/HI) template generation via Groq LLaMA-3.3-70B
  - Self-learning loop — classifies templates on live CTR/engagement, suppresses underperformers, injects high-CTR examples as LLM few-shot context; auto-updates timing & frequency in Iteration 1
- **Minimum-Variance Portfolio Optimizer** *Dec. 2025 – Present*
  Python, Pandas, NumPy, yfinance, PyPortfolioOpt, Modern Portfolio Theory Github
  - Quantitative pipeline for lowest-risk long-only portfolios; applied Ledoit-Wolf shrinkage and nearest-PSD correction for covariance stability; enforced no-short-selling constraints with weight normalization
- **Kelly Criterion for Fixed-Limit Games** *Dec. 2025 – Jan. 2026*
  Python, Game Theory, Stochastic Modeling LinkedIn
  - Derived the **Stochastic Gate** to convert optimal Kelly fraction f^* into a Kelly Frequency for fixed-bet games; built a Python decision engine matching the optimal geometric growth curve under hard bet-size constraints

Technical Skills

- **Programming**: C++, Python, C
- **Libraries & Tools**: PyTorch, NumPy, Pandas, Scikit-learn, UMAP, Matplotlib, yfinance, PyPortfolioOpt, LangChain, Git, LaTeX
- **Key Concepts**: Deep Learning, Transformer Architectures, Stochastic Calculus, Modern Portfolio Theory, Data Structures & Algorithms, Game Theory, Risk Modeling

Positions of Responsibility

- **Associate**, Consulting & Analytics Club, IIT Guwahati *Dec. 2025 - Present*
- **Associate**, Udgam, IIT Guwahati (E-Cell) *Oct. 2025 - Jan. 2026*

Achievements

- **JEE Advanced 2025**, All India Rank 1941 | Top 1% among 1.5 Lakh+ candidates *2025*
- **JEE Mains 2025**, All India Rank 4738 | 99.69 percentile *2025*
- **Competitive Programming**, Codeforces Specialist (Max Rating: 1437) | CodeChef 2-Star | 200+ problems solved *2025*